

their primary coding languages.

- **Tizen uses TouchWiz, like Android does:** Most potential users of Tizen have, so far, been comfortable with Android or iOS systems. The latest version of Tizen doesn't look very different from Android, which is TouchWiz-layered. Like Android phones, Tizen also has a swipe-down notification bar with a toggle bar, as used in the TouchWiz UI, and it supports multi-tasking, power saving, firewalls to block malicious sites and other features.
- **Samsung Tizen apps:** Apps have always played a lead role in the success of any operating system. Tizen is supposed to support all the native and Web apps, as Bada apps have also been incorporated into this system. A Tizen app challenge took place in 2013 when Samsung awarded US\$ 4 million to 64 winners who made native Tizen apps and also some HTML5-based apps.

Tizen's reach across the electronics industry

The Tizen OS is based on the Linux kernel and GNU C Library, through the Linux API. It's a cross-platform OS for smartphones, tablets, in-vehicle infotainment systems, smart TVs, PCs, smart cameras, wearable gadgets, Blu-ray players, printers and smart home appliances. Samsung's NX300M smart camera, released in October 2013, was the first consumer product based on Tizen. Then the first Tizen tablet from Systena was shipped and, later, Samsung's Gear 2 smartwatch was launched, which also runs on Tizen.

The challenges Tizen has faced in the industry

Last year, Samsung tried to launch a Tizen smartphone several times in Japan, Russia and India. But due to lack of support from the market, including carriers and app developers, the plans never materialised. But, finally, Samsung has overcome all the barriers. Its executives are of the opinion that if the new smartphone is able to garner handsome profits in the Indian market, then it would create a positive impact on the Tizen build and more app developers will be attracted towards the OS. Throughout 2014, Samsung struggled with the Tizen smartphone, and in the meantime, some low-cost competitors like Micromax and Xiaomi ate into Samsung's share of the Indian market by launching several budget smartphones. Samsung has, so far, contributed the most to the growth of Android, and has never been too aggressive about the launch of its Tizen OS.

... and the Tizen-based Samsung Z1 smartphone finally arrives

Finally, the Tizen-powered Samsung Z1 smartphone is up for sale in India with a price tag of Rs 5,700. The new smartphone offers faster booting times and long-lasting battery life. Samsung decided to launch the Tizen smartphone in India as it's a cost-sensitive market. The Z1 is a low-end entry-level smartphone featuring a 10.16-

cm (4-inch) WVGA screen, 3G connectivity, a 1.2GHz dual-core processor, 768MB RAM, a 1500mAh battery, a 3.1-megapixel rear camera and 4GB of storage (along with a microSD slot). Additionally, Samsung Z1 will also offer consumers free access to content on Club Samsung and a live music streaming experience on Hungama.com. Users can also access TV shows and over 7,000 Bollywood and Hollywood movies for a fee of Rs 1,750 per annum, since a deal has been struck between Samsung and mobile live TV services providers, nexGTV and BoxTV. The smartphone is available in white, black and wine red colour variants.

Tizen vs Android: The future

Initially, there were plans for Tizen to be installed in Samsung's high-end Galaxy devices, but that strategy was subsequently changed, and Tizen became a part of Samsung's budget product line-up. The strategy might have changed because the company believed that the OS would do better in the budget product market in the future. Mozilla also released its Firefox OS for phones, keeping these emerging markets in mind, as these markets have remained unexplored in the global scenario. So if Samsung's strategy to use Tizen to capture the emerging budget market succeeds, then the OS might get promoted to its higher-end devices too.

But, currently, Android is all-pervasive across all kinds of smartphone and tablet markets. So, Samsung will have to face serious challenges in persuading developers to create content for Tizen-based smartphones and tablets. However, Tizen's future lies in the IoT sector, which is still in its infancy and is a space in which Android has not yet become dominant. Tizen may offer some good alternatives to the Android-based IoT services, but winning over Android fans will not be so easy for this new OS.

As Tizen is open source software, a developer can use the Tizen Common Base to create a new device based on the operating system. The Tizen project is governed by a Technical Steering Group, based within the Linux Foundation. The industry role of Tizen is looked after by the Tizen Association. Tizen is going to be a crucial part of Samsung's Internet of Things strategy, which will connect devices and appliances, making our lives smarter. As of now, Tizen is far from toppling the Android operating system, but keeping in mind its positioning and promotions, Tizen does seem like its heading towards a bright future. The launch of the Samsung Z1 smartphone, based on the Tizen OS, is a clear signal that Samsung is geared up to compete with Google in the mobile software segment now. 

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